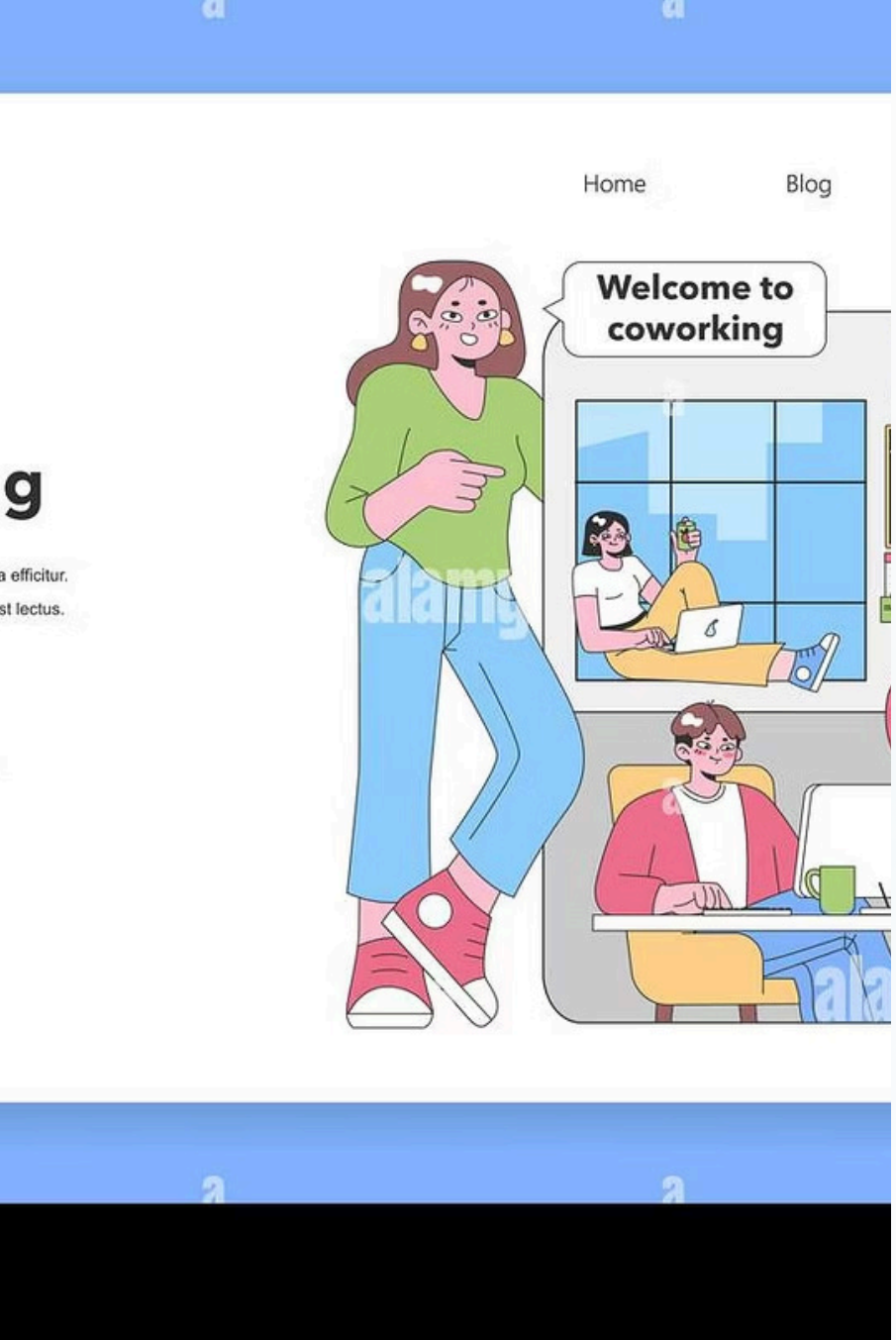


# MERN Accommodation Listing Platform

Full-stack application enabling browsing, creation, management and review of accommodation listings. Built with the MERN stack and RESTful APIs for maintainability and scale.

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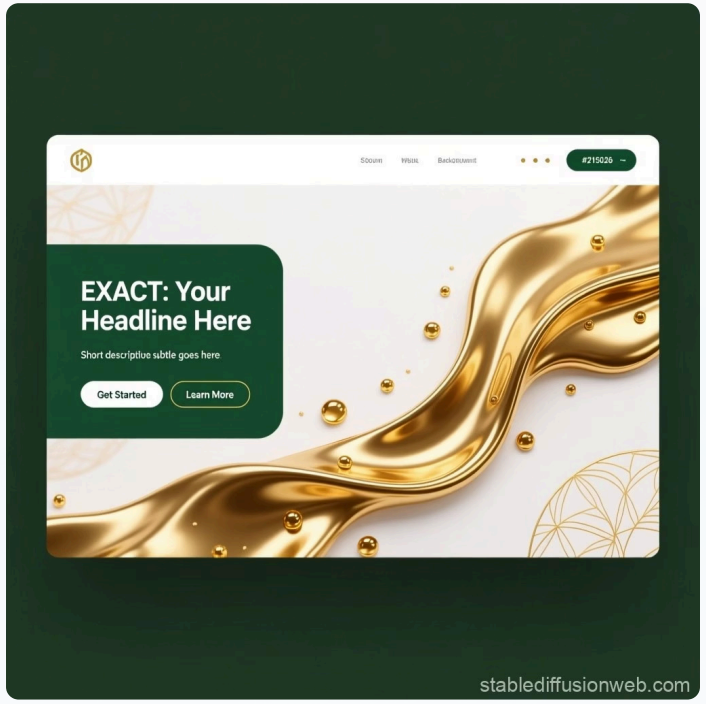
# Project Snapshot




**Purpose:** Provide a secure, user-friendly marketplace for hosts and guests.

**Core objectives:**

- Secure authentication & session management
- Full CRUD for property listings
- Structured reviews & ratings
- Scalable, RESTful API architecture

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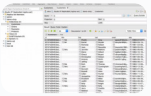
# Technology Stack

## Frontend

HTML, CSS, JavaScript and EJS for server-rendered dynamic templates and fast SSR experiences.

## Backend

Node.js and Express.js powering RESTful controllers, middleware and routing.



## Database

MongoDB via Mongoose for flexible schemas and references (users, listings, reviews).

## Auth & Security

Passport.js, express-session + MongoStore, password hashing and protected routes.

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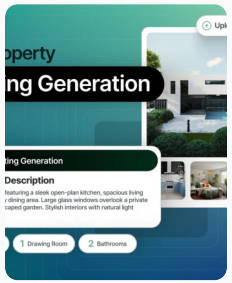
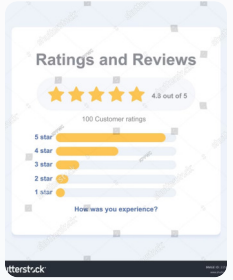
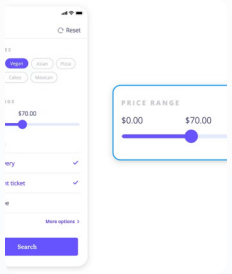
# Key Features — UX & Functionality




## User Authentication

Secure registration and login, persistent sessions stored in MongoDB, and session invalidation on logout.

## Property Listings

Create/edit/delete listings with title, description, price, location and image set. Owner-only edit/delete permissions enforced server-side.

## Reviews & Ratings

Users add ratings and comments; authors can delete own reviews. Ratings aggregated per listing for transparency.

## Filtering & Categorisation

Category tags and server-side filters rendered with EJS for fast, SEO-friendly listing pages.

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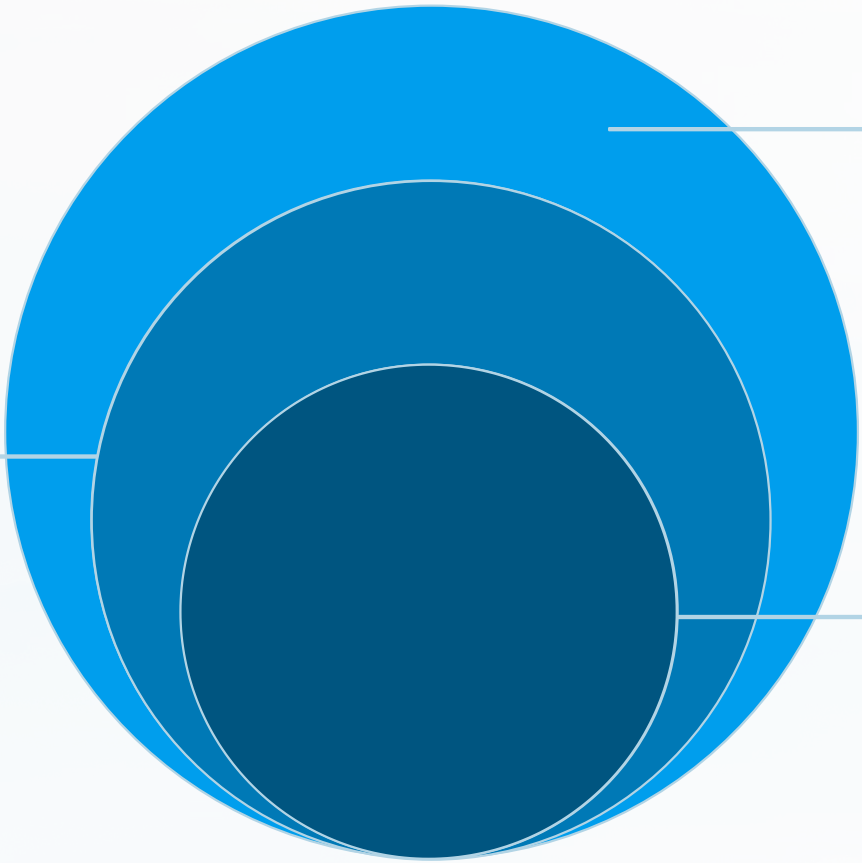
# System Architecture



### Database Layer

MongoDB persistence via Mongoose schemas

### Server Layer

Node.js + Express controllers and middleware

### Client Layer

EJS views in browser, sends REST requests

The client communicates with Express REST endpoints which encapsulate controllers and middleware; persistence is handled by Mongoose models in MongoDB. Separation of concerns enables scaling and focused testing.

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# Database Schema Overview

## Embedding in MongoDB

**Users**

- ID
- First name
- Last name
- Department
- Role
- Location X
- Location Y

```
{
  "id": "1",
  "firstName": "John",
  "lastName": "Doe",
  "locationX": 45.123,
  "locationY": 47.232,
  "department": "Development",
  "role": "Frontend Developer"
}
```

### User Schema

username, email, hashedPassword, createdListings (ObjectId[])

### Listing Schema

title, description, price, location, owner (ObjectId ref User), images[], reviews[]

### Review Schema

rating, comment, author (ObjectId ref User), createdAt

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# Security & Auth Details

- **Password hashing:** bcrypt with strong salt rounds.
- **Session management:** express-session + connect-mongo (MongoStore) for persistence across restarts.
- **Authentication:** Passport.js local strategy; middleware protects create/edit/delete routes.
- **Authorization:** Role/ownership checks in controllers to ensure only owners can modify resources.
- **Validation:** Input sanitisation and server-side validation middleware to prevent injection and malformed data.

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# API Design & Patterns

01

## Resourceful routing

/listings, /listings/:id, /listings/:id/reviews — consistent CRUD verbs (GET, POST, PUT, DELETE)

02

## Controller separation

Thin routes, fat controllers: business logic isolated for testability and reuse.

03

## Middleware

Auth, validation, error handling and file upload middleware (placeholder for cloud upload integration).

04

## Pagination & filtering

Query params for page, limit, sort, price range and category to keep responses lean and performant.

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# Challenges & Solutions

## Auth & sessions

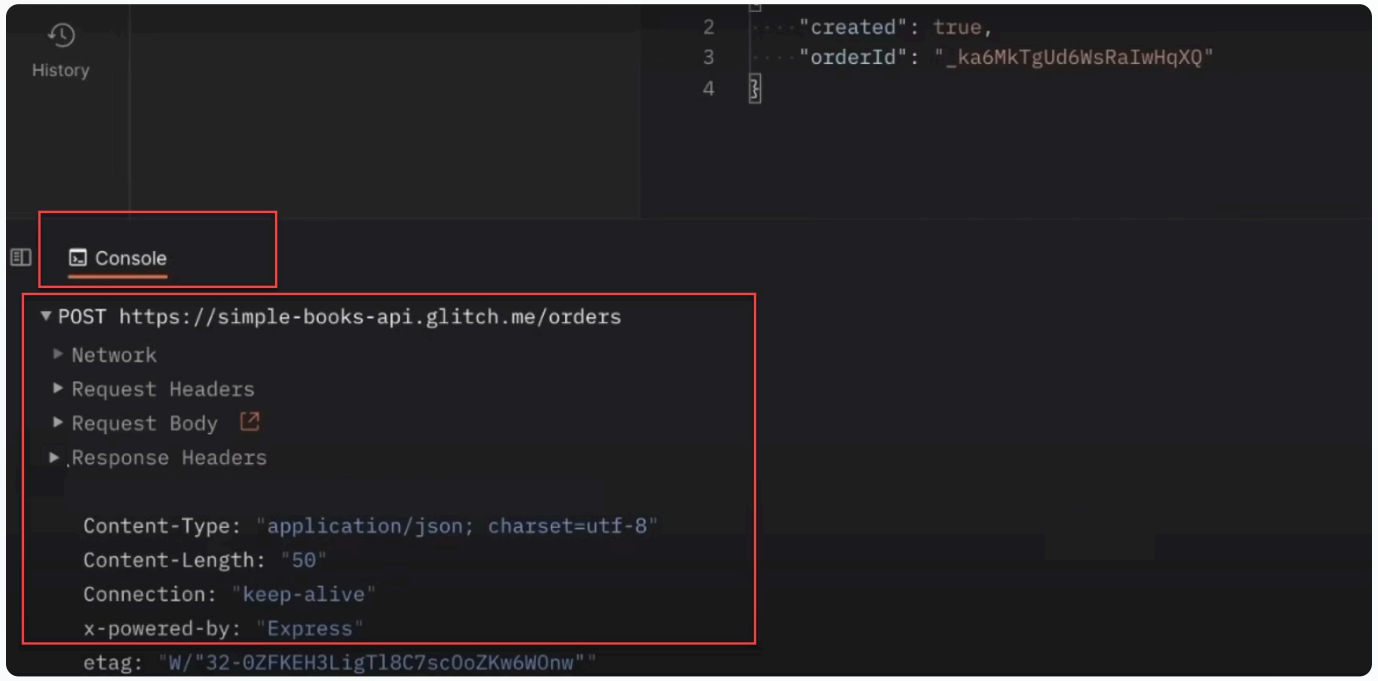
Handled with Passport.js and MongoStore; implemented session rotation and logout invalidation.

## Data relations in MongoDB

Used references with population for listings → reviews → author, balancing embeds vs refs for scale.

## Authorization logic

Ownership middleware centralised to prevent repeated checks and reduce bugs.

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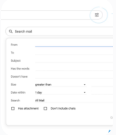
# Roadmap & Next Steps

### Payments

Integrate Stripe/PayPal for booking deposits and host payouts; secure PCI-compliant flow.

### Cloud image uploads

Move file storage to S3 or Cloudinary with presigned uploads and CDN delivery for performance.



### Advanced search

Full-text search, geospatial queries, faceted filters and relevance ranking.

### Admin & deployment

Admin dashboard for moderation and analytics; deploy to scalable hosting (containers, managed MongoDB, CI/CD).

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**Conclusion:** The platform demonstrates practical MERN full-stack skills — secure authentication, expressive data models, and RESTful architecture — and is positioned for incremental enhancements toward a production system.

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